$$\mathbf{LM}|\lambda, \mu\rangle = \lambda\mu|\lambda, \mu\rangle.$$

The eigenvalues λ, μ are of course just numbers and it doesn't matter which one appears first when we multiply them. Thus, if we reverse the order of these operators, and let the operator **ML** act on the same basis vector, we get the same result:

$$\mathbf{LM}|\lambda, \mu\rangle = \mathbf{ML}|\lambda, \mu\rangle,$$

or, more succinctly,

$$[\mathbf{L}, \mathbf{M}]\ |\lambda, \mu\rangle = 0, \qquad (5.1)$$

where the right-hand side represents the zero vector. This result would not be very helpful if it were only true for a particular basis vector. But the reasoning that leads us to Eq. 5.1 is valid for *any* of the basis vectors. That's enough to ensure that the operator [**L**, **M**] = 0. If an operator *annihilates* every member of a basis, it must also annihilate every vector in the vector space.[1] An operator that annihilates every vector is exactly what we mean by the *zero operator*. Thus, we prove that if there is a complete basis of simultaneous eigenvectors of two observables, the two observables must commute. It turns out that the converse of this theorem is also true: if two observables commute, then there is a complete basis of simultaneous eigenvectors of the two observables. To put it simply, the condition for two observables to be simultaneously measurable is that they commute.

As we mentioned earlier, this theorem is more general. One may need to specify a larger number of observables to completely label a basis. Regardless of the number of observables that are needed, they must all commute among themselves. We call this collection a *complete set of commuting observables*.

5.1.2 Wave Functions